

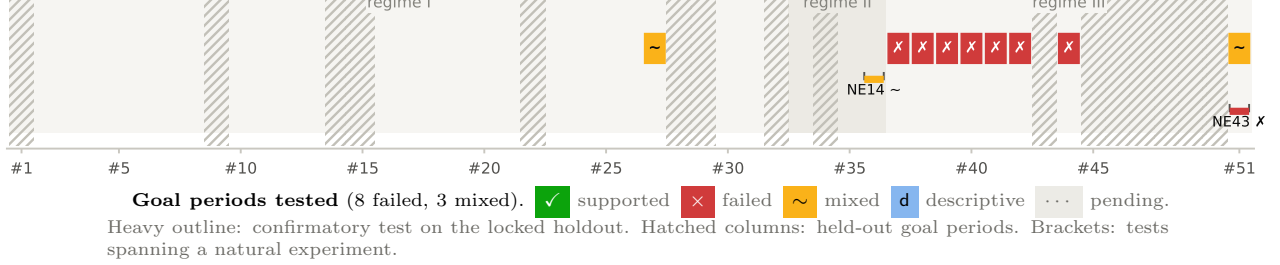
H14: Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

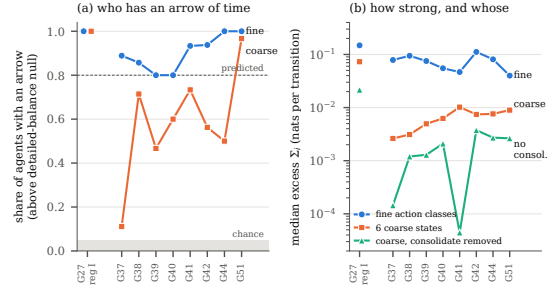
Bottom line. Every agent's fine tool workflow is irreversible without the scaffold; the coarse arrow and the regime contrast are scaffold, and no family, collective or useful-work arrow appears.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H14 tests whether each agent's behavior is irreversible. It also tests whether each model family has its own arrow of time (HH19), and whether the swarm is more irreversible than its agents (HH67). Round 1 covers 9 periods (#27 in regime I; #37–#44 and #51 in regime III). On 6 coarse states the arrows are small (0.002–0.010 nats/transition), and the scaffold's consolidation step makes most of them. Regime I gives 7–29× larger arrows. No family or collective term appears. Round 1b uses shell sub-classes from the fixed **bash.head**, an agent-only chain with a burn-in after scaffold resets, and Jev v3.1 semantic states. Every agent's fine arrow is now irreversible (9/9 periods), and it survives scaffold removal (41–104% kept). The regime contrast vanishes on every corrected chain. Semantic states carry a small pooled arrow (7/9) that does not track output. The natives (NE43, NE14, loops against progress) give no support. The holdout has not run.



Observables. (a) The share of agents whose state sequence beats a detailed-balance surrogate (same occupancies, symmetric transition counts, no currents). Almost all agents do on fine action classes. On the 6 coarse states only a minority do, except in the long #51 window and in regime I (#27). (b) The median excess irreversibility. Fine micro-workflow carries about 10× the coarse arrow. Most of the coarse arrow vanishes when the scaffold's consolidation step is removed.

PHYSICAL MODEL

Kinetic Potts (models 02, 10). $x_i(t) \in \{1..q\}$ is agent i 's behavior state per turn (single agent) or per 1-min bin (swarm). P_{ab} and π_a are its transition matrix and occupancies. Σ_i is the entropy production per transition: the pair KL, a lower bound for non-Markov agents. The Aguilera–Ito–Kolchinsky dual estimates it on antisymmetrized transition indicators g_{ab} , cross-fitted by day (Newton step $\hat{\Sigma}_N$). J_{ij}^{ab} are cross-agent couplings. $\Delta\Sigma$ is the collective increase of the bound, and A_{abc} is a cycle affinity.

$$\begin{aligned}\Sigma_i &= \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, & g_{ab} &= \mathbf{1}[a \rightarrow b] - \mathbf{1}[b \rightarrow a] \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle], & \hat{\Sigma}_N &= 2 \langle \bar{g}_a^\top K^{-1} \bar{g}_b \rangle_{a \neq b} \\ P(x'_i = a | \mathbf{x}) &\propto e^{h_i^a + \sum_j J_{ij}^{a, x_j}}, & \Delta\Sigma &= \Sigma_{1+2} - \Sigma_1 \\ A_{abc} &= \ln \frac{P_{ab} P_{bc} P_{ca}}{P_{ac} P_{cb} P_{ba}}\end{aligned}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

This is mostly a careful measurement of the harness, with one agent-level residue. The coarse arrow and the regime-I/III contrast are scaffold structure: sessions and consolidation. Fine tool workflow (e.g. read → edit → commit) survives scaffold removal and is irreversible in every agent. It is a per-agent fingerprint, not a steering handle. Semantic irreversibility is real but small, and it does not track output (ρ from -0.34 to $+0.14$). EP is therefore not a useful-work meter. For EP on agent logs, use a calibrated detailed-balance or block-flip null, and subtract the null before you compare agents. Never apply a reset burn-in on coarse windows: it manufactures arrows. An operator would change nothing.

H14-behavior-entropy-production · generated 2026-10-04 12:01 by infra/summaries/build_summaries.py · card hypotheses/H14-behavior-entropy-production/README.md · ratings: H14 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
fine (shell sub-classes)	100% of agents in 9/9	holds
$\geq 80\%$ irreversible		
coarse arrow mostly	5/8 keep $< 50\%$ after removal	failed (narrow)
scaffold ($\geq 6/8$)		
fine arrow survives	kept 41–104%; 75–100% of holds	
scaffold removal	agents	
pooled Jev v3 arrow	$p \leq 0.005$; 7/9 periods	holds
(G38, G51)		
regime I $\geq 2\times$	(v3 ratios 0.4–4.4	failed
states)		
lab η^2 (fine, scaffold-free)	p 0.35; Anthropic $>$ OpenAI 8/8, p 0.04	failed / fragile
v3 EP tracks output	ρ +0.14, -0.34, -0.16	failed
N1 NE43 inside placebo range	3/4 changes inside	failed (marg.)
N2 NE14: coarse drops more than v3	0.25 vs 0.72 (CIs span 1)	mixed
N3 productive $>$ stuck EP	0/3	failed

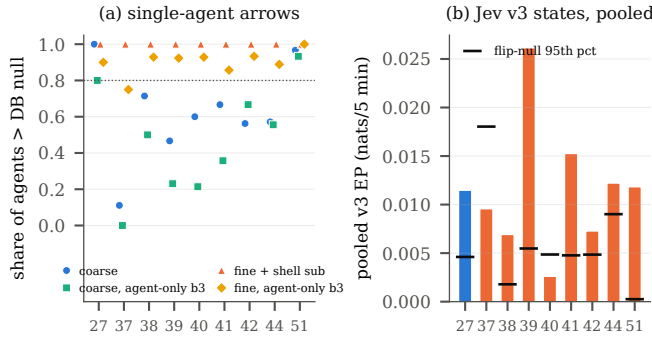
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Soft-label observables attenuate EP by roughly the square of label accuracy. Per-agent v3 tests are therefore nearly blind, and only pooled tests are primary. The pre-registered reset burn-in on 5-min windows is a selection artifact, because resets start activity bursts; it is reported as invalid. The Newton step blows up for a few strongly asymmetric agents (one G40 value is 16.5). That agent drives the fragile Anthropic $>$ OpenAI result. NE14 has one pre-change day, and NE43 windows are 5 days. Non-Markov agents make all values lower bounds.



Second figure. (a) The share of agents that beat the detailed-balance null, per period. Fine classes with shell sub-classes (triangles) are irreversible for every agent and stay so without the scaffold (diamonds). The coarse arrow (dots) mostly does not survive removal (squares). (b) Pooled Jev v3 entropy production per 5-min transition (bars) and the block-flip null's 95th percentile. Regime I (#27, blue) is not above regime III.

GOAL PERIODS IN DETAIL

G27 $\sim$ 1b: mixed (unchanged: the pre-registered coarse chain...
G37 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G38 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G39 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G40 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G41 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G42 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G44 $\times$ 1b: failed (unchanged: the pre-registered coarse chain...
G51 $\sim$ 1b: mixed (unchanged: the pre-registered coarse chain...
NE14 $\sim$ mixed
+1 more periods: see the card.

ROUND 2

Measure count-matched within-agent EP across goal kickoffs (H56's design) on `act.sh`. Denoise semantic states with an HMM before EP.

- **Essence.** The arrow of time of work: progress is irreversible, chatter is reversible.
- **H14-R1** Entropy production on work states (plan, build, verify, ship; Jev) correlates with output: useful irreversibility.
- **H14-R2** Loops are near-reversible cycles with zero progress; productive phases produce entropy.
- **H14-R3** The scaffold contributes a fixed entropy-production floor (the consolidation clock); subtracting it leaves agent-generated irreversibility (E1).